



Probability with Engineering Applications: Week5

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Programme

- Quantiles
- Poisson process
- Applications of exponential distributions

Quantiles

Quantiles

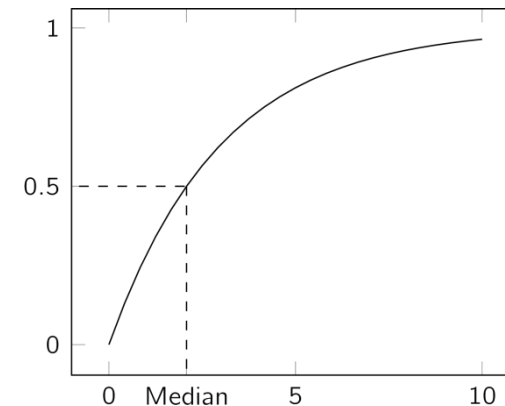
Definition:

Let X be a continuous random variable and $0 \leq p \leq 1$. The p th quantile or $100p$ th percentile of the distribution of X is the smallest number q_p such that

$$F_X(q_p) = P(X \leq q_p) = p.$$

Example:

The median of a distribution is its 50th percentile.



Determine the

- 0.5th quantile of a $U(1, 3)$ distribution
- 80th percentile of an $\text{Exp}(\lambda)$ distribution
- 10th percentile of a standard normal distribution.

Standard Normal Distribution Table (Right-Tail Probabilities)

[illegible]

Distribution properties

Consider the following statements:

1. Let $X \sim \text{Exp}(\lambda)$. Then $P(X > x)$ increases as λ increases.
2. Let $X \sim N(\mu, \sigma^2)$. Then always $P(X < \mu) = \frac{1}{2}$.

Which of these two statements is/are true?

- A.** None
- B.** Only 1
- C.** Only 2
- D.** Both

Distribution functions

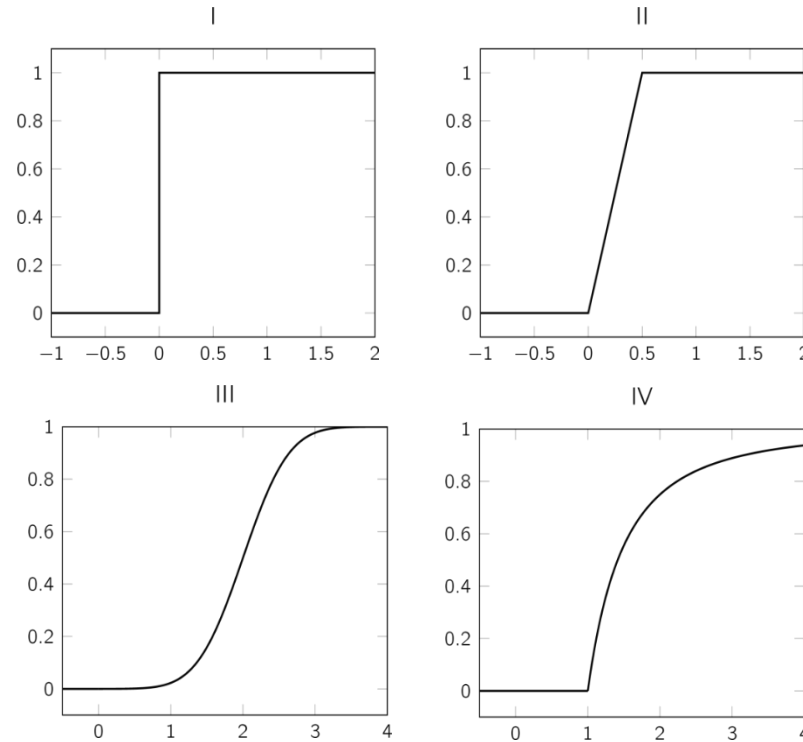
Consider the following four graphs below. Which corresponds to the distribution function of a uniform random variable and which to the distribution function of a Pareto distribution?

A. Unif: I, Par: III

B. Unif: II, Par: III

C. Unif: I, Par: IV

D. Unif: II, Par: IV



Poisson process



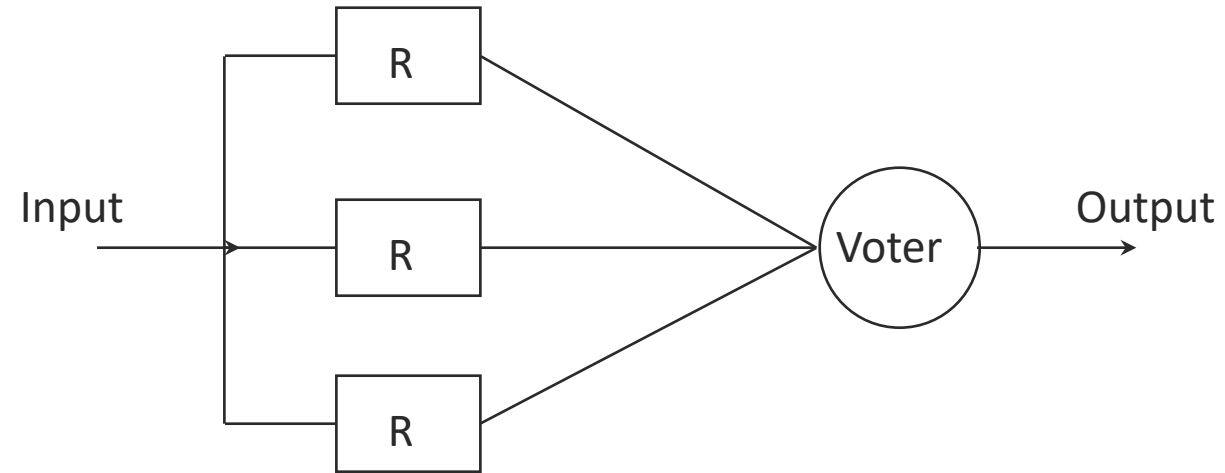
Poisson process

More on exponential distribution

Example 1 (Series system)



Example 2 (TMR system)

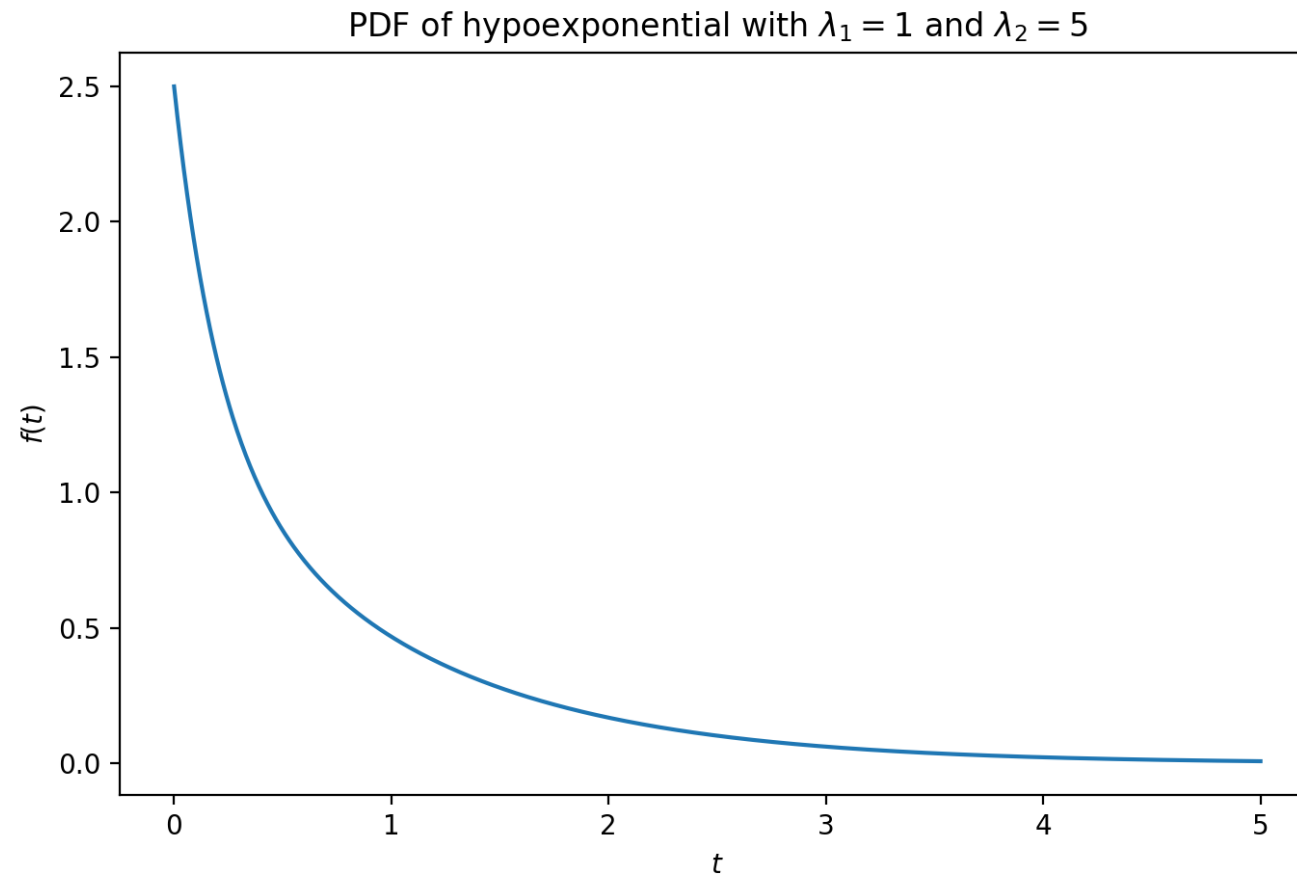


Example 3 (Production system)

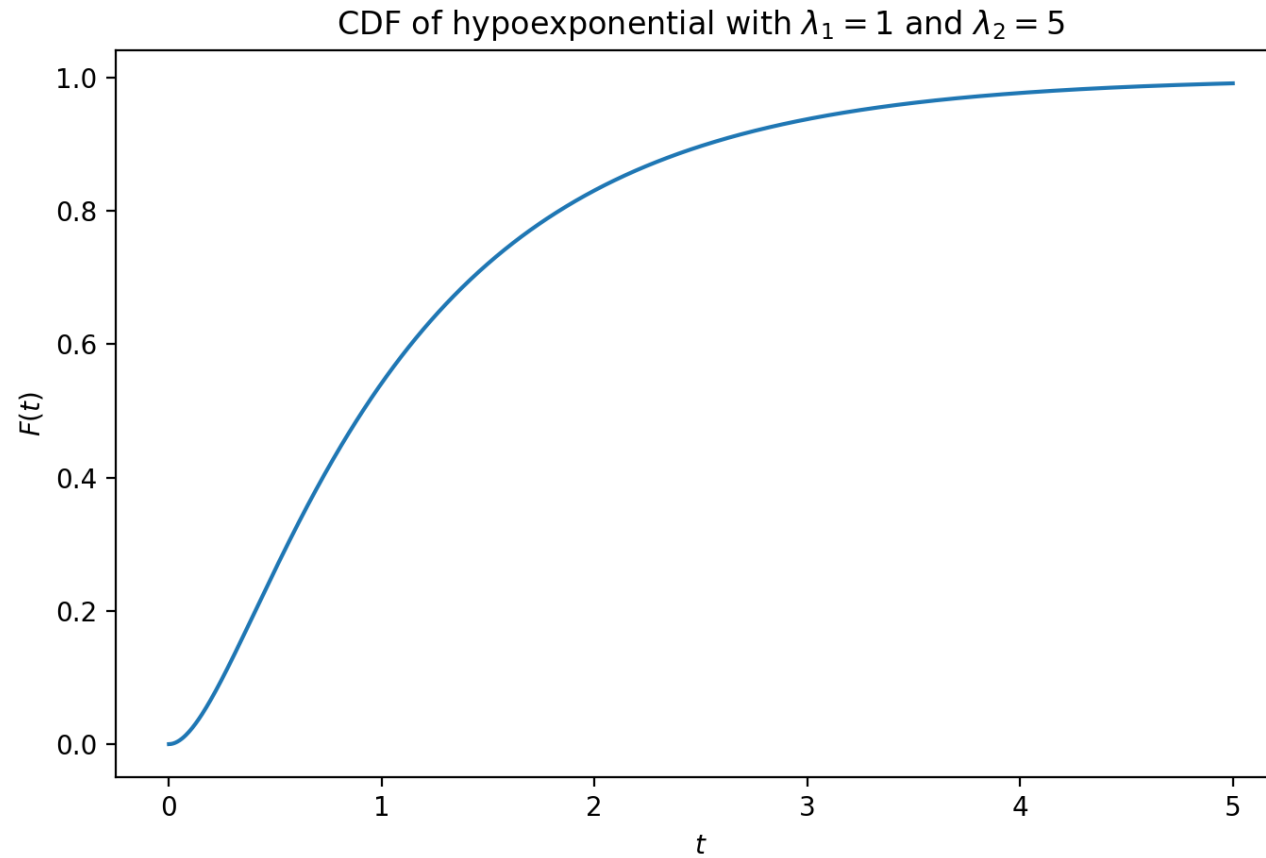
- Many processes in nature can be divided into sequential phases.



The PDF of the Hypoexponential Distribution



The CDF of the Hypoexponential Distribution





See you next week